Gravitational Field

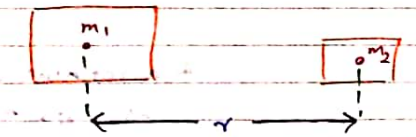
1- Every object attracts every other object

2- Force of attraction b/w two objects is called "Gravitional force"

e.g. Moon & Earth
Earth & Sun
Marker to Marker

Newtons Law of Gravitation

Force of attraction b/w two objects is directly propotional to product of masses of object and is inversly propotional to square of distance b/w centres of object



(distance b/w centre of 2 masses)

$F \propto m_1 m_2$

$F \propto \frac{1}{r^2}$

$F \propto \frac{m_1 m_2}{r^2}$

$F = -(\text{constant}) \frac{m_1 m_2}{r^2}$

$$F = \frac{-G m_1 m_2}{r^2}$$

i- G = Universal Gravitional Constant
$= 6.67 \times 10^{-11} Nm^2/kg^2$

* Universal cz it remains same all over the universe

ii- -ve sign => Attractive force

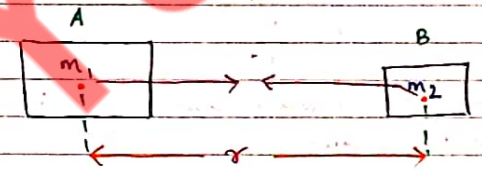
* +ve sign => Repulsive force
+ve -ve sign => Both attraction & repulsion

Nature of Force
Attractive Repulsive

Note

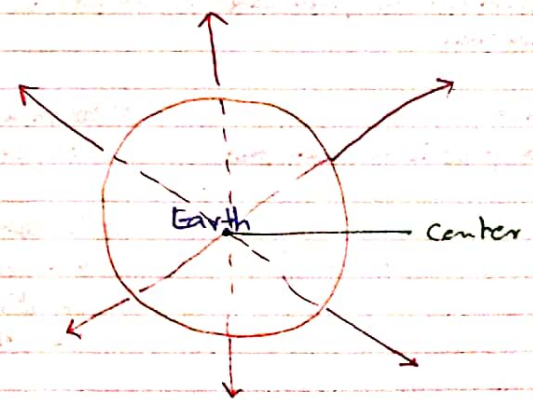
- Force of attraction b/w two charges is "electrostatic force"
- Force of attraction b/w two magnets is "magnetic force"

iii-



* The distance "r" is always taken from the centre of each masses
* We assume that all the mass/weight of the body acts/appears to act at its centre that's why distance is r and we draw gravitional field from the centre of object

To prove this



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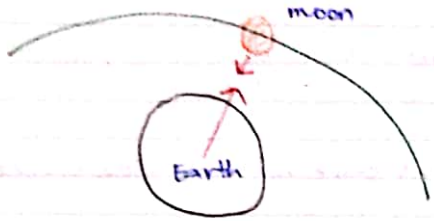
iv.



→ Gravitational force acts along the line joining the centres of the two objects
↳ Helps to label the diagram

* Earth is larger and moon is smaller
* Both exert the same force on each other
* Force is dependant on a) $m_1 m_2$
b) $\frac{1}{r^2}$

$F = \frac{m_1 m_2}{r^2}$

as $m_1 m_2$ and r^2 remains same thus F remains or is same

* Moon has a smaller mass so the force is sufficient to produce motion in it as it revolves round the earth

v. Exists in Pair
→ Gravitional force exists in pair that is whenever two objects exert a force on each other, they exert equal force on each other opposite in direction

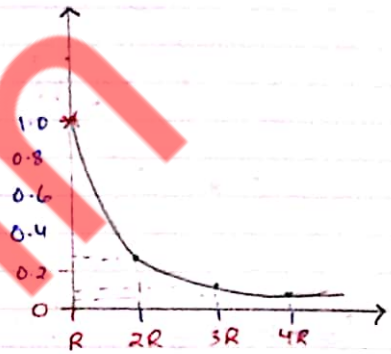
vi. Weak force
→ Gravitational force is a weak force

* For small objects its very negligible
* It becomes noticible when one of the object's mass equals that of earth

vii. Inverse square law

$F \propto \frac{1}{r^2}$

→ If d 2x then Force 4 times decrease
→ If d 3x then Force 9 times decreases
→ If d 4x then force 16 times decreases



Inverse Square Law Graph

* Curved graph
* Never touches x-axis
* Line with negative gradient

Earth

→ Distance b/w surface of Earth from centre of Earth is "radius"

→ If moved away from the surface of Earth

for Earth
↳ Relation b/w F & r

viii.

$R = F_G$

$W = F_G$

W = weight

* At pole, with respect to rotation of earth no circular motion
↓
No centripetal force (No R.F) resultant force

* Earth is a perfect sphere — assumed
* At both equator + pole the pull of earth will be same.
* Radius of earth is same for at pole & at equator

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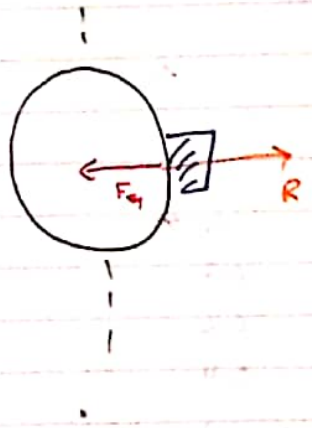


At equator, with respect to the rotation of earth perform circular motion

↓

centripetal force acts

↓

Resultant force present

$F_G - R = F_C$ — Centripetal force

$F_C = F_G - W$

$$\boxed{W = F_G - F_C}$$

for equator of Earth

Note

Weight is more at pole than at equator

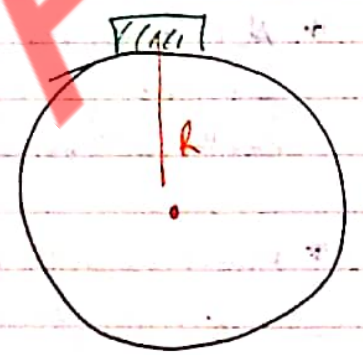
$W = mg$

At pole & equator the g - acceleration free fall would be different

Note

At equator, circular motion is done by object so it shows for F_G and rest as F_C.

Relation b/w G and g



mass of Earth = M

mass of object = m

Radius of Earth = R

$F_G = \frac{GMm}{r^2}$ — i

$W = mg$ — ii

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$W = F_G$

$$\cancel{m}g = \frac{GM\cancel{m}}{R^2}$$

$$\boxed{GM = r^2 g}$$

$$\boxed{M = \frac{R^2 g}{G}}$$

$R = 6.4 \times 10^6\ m$

$G = 6.67 \times 10^{-11}\ Nm^2/kg^2$

$g = 9.81\ ms^{-2}$

$M = 6.0 \times 10^{24}\ kg$

$$\boxed{V = \frac{4}{3}\pi R^3}$$

$$\boxed{\rho = \frac{M}{V}}$$

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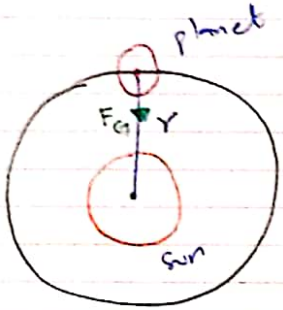
Motion of a planet around the sun



Mass of sun = M

Mass of Planet = m

Radius of orbit = r

Angular Velocity = ω

Linear Velocity = v

Time Period = T

$F_G = \frac{GMm}{r^2}$ — i (Pull of sun on planet)

$F_c = mr\omega^2$ — ii (Centripetal force on planet)

* Gravitional pull of sun will provide centripetal force on planet

$$\boxed{F_c = F_G}$$

* Both -ve sign cancel each out each other

$$\cancel{m}r\omega^2 = \frac{GM\cancel{m}}{r^2}$$

$$\omega^2 = \frac{GM}{r^3}$$

$$\left(\frac{2\pi}{T}\right)^2 = \frac{GM}{r^3}$$

$$\frac{4\pi^2}{T^2} = \frac{GM}{r^3}$$

$$\frac{T^2}{4\pi^2} = \frac{r^3}{GM}$$

$$T^2 = \underbrace{\boxed{\left(\frac{4\pi^2}{GM}\right)}}_{\text{constant}} r^3$$

$$T^2 \propto r^3$$

Kepler's Law

Square of time period of a planet is directly propotional to the cube of radius of orbit of that planet

Derivations

$$\boxed{\omega = \sqrt{\frac{GM}{r^3}}}$$ — start from $F_c = F_G$

$$\frac{2\pi}{T} = \sqrt{\frac{GM}{r^3}}$$

$$\boxed{T = 2\pi\sqrt{\frac{r^3}{GM}}}$$

$$v = r\omega$$

$$v = r \times \sqrt{\frac{GM}{r^3}}$$

$$v = \sqrt{\frac{GM \times r^2}{r^3}}$$

$$\boxed{v = \sqrt{\frac{GM}{r}}}$$

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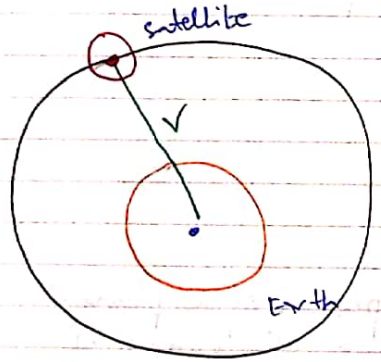
Motion of a Satellite around the Earth



mass of Earth
mass of satellite
radius of orbit
angular velocity

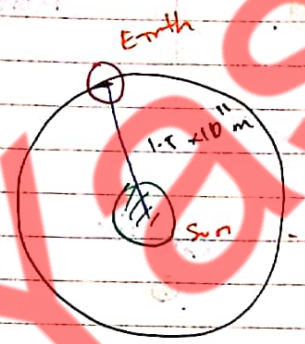
(mass / object)

* Anything revolving around sun is called a "Planet"

* Anything revolving round earth is called it's "satelite"

↳ Moon is earth's natural satellite

Q.



a) $\omega = ?$

b) $v = ?$

c) $M = ?$

$G = 6.67 \times 10^{-11}\ Nm^2/kg^2$

a) $\boxed{\omega = \sqrt{\frac{GM}{r^3}}}$ $\Rightarrow$ $\omega = \frac{2\pi}{T}$ — 365 days

$= \frac{2 \times 3.142}{31536000}$

$= 1.9 \approx 2.0 \times 10^{-7}\ ms^{-1}$

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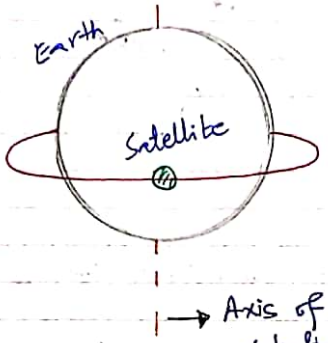
2-8-21 Gravitational Field Monday

Properties



1- Time period of circular motion of satellite is same as that of rotation of earth i.e $\boxed{24 \text{ hours}}$

2- Direction of Circular motion of satellite is same as that of rotation of Earth i.e $\boxed{\text{west to East}}$

→ Justification

- Sun appears like this
  ↓
  It's stationary ∴ Earth rotates

* We are considering the rotational motion of Earth & circular motion of satellite

3- Axis of Circular Motion of satellite is same that of rotation of Earth.

→ The effect of these 3 properties makes the satellite stationary.

* If any of these 3 properties were not fulfilled the satellite wouldn't have been stationary.

Geo - Stationary Satellite

A satellite which always remains at the same place above the surface of Earth.

OR

A satellite which appears stationary with respect to rotation of Earth.

This is the circular motion of any satellite.

$$\boxed{T = 2\pi\sqrt{\frac{r^3}{GM}}}$$

T = Time period of circular Motion of satellite
r = Radius of orbit
G = Universal gravitational constant
M = Mass of earth

* Not all satellite are geo stationary

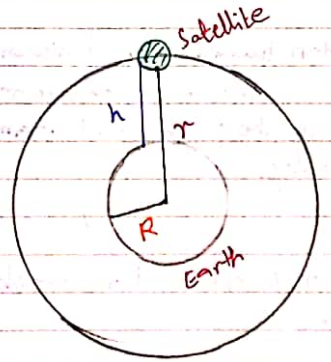
$G = 6.67 \times 10^{-11}\ Nm^2/kg^2$
$M = 6.0 \times 10^{24}\ kg$
$R = 6.4 \times 10^6\ m$

$\therefore \boxed{r = 4.2 \times 10^7\ m}$

=> In this expression, if used for geo-stationary satellite — there will be only one unknown i.e "r" cz T = 24 hours

* All "r" are constant for geo stationary satellite because all other quantities are constant.

* All "h" are also same as "R" and "r" are same



R = Radius of Earth

$\boxed{r = R + h}$

$\boxed{h = r - R}$

$h = 3.6 \times 10^7\ m$

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Justification of the following

1- h – same, so launch the satellite from earth to distance "h" ∴ same time period

2- West to East – To prove this, start the engine of satellite at h so that circular motion starts.

3- Launch satellite from equator

Note

- Satellite (geo-stationary) can only be acheived if we launch the satellite from equator.

Geostationary Satellite

Uses

1- Telecommunication
2- Satellite channels
3- Spy satellite
4- weather forecasting

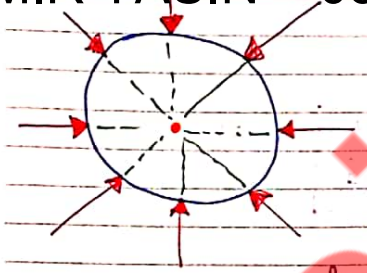
Transmitter → receiver → set direction towards geo-stationary satellite. This only has to be done once the signal will transmit continously.

Gravitational Field

A region around an object within which it can exert a force on a unit mass.

Gravitational Field line

Path followed by a unit mass within the gravitational field is called "Gravitational field line"



Representation of earth's gravitational field pattern.

→ Gravitational field lines concentrate at mass

→ Here all gravitational lines concentrate at centre, so it's mass lies at centre

→ For bigger objects we assume that all it's mass is concentrate at the centre.

Property of gravitational field

Gravitational Field Strength

Force acting on a unit mass at a point inside gravitational field is called "gravitational field strength" at the point.

* gravitational field strength $= \frac{\text{force}}{\text{mass}}$

∴ $\boxed{g = \frac{F}{m}}$

Earth

$F = \frac{-GMm}{r^2}$ (Newtons law of gravitation)

$\boxed{g = \frac{F}{m}}$

$\boxed{g = \frac{-GM}{r^2}}$

r = distance b/w centre point you're measuring the gravitational field strength

M = Mass of Earth → m will be cut out

G = universal constant

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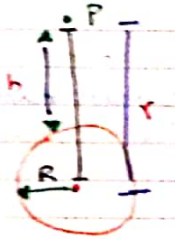
Case I

At surface of earth

$$g' = \frac{-GM}{R^2} \rightarrow (i)$$

Case II

At a point above the surface of earth



$r = R + h$

$$g'' = \frac{-GM}{r^2} \rightarrow (ii)$$

$$g'' = \frac{-GM}{(R+h)^2} \rightarrow (iii)$$

* Eqn #2 ÷ Eqn #1

$$\frac{g''}{g'} = \frac{R^2}{r^2}$$

$$g'' = \frac{R^2}{r^2} \times g' \rightarrow (iv)$$

Q.

$g' = 9.81\ Nkg^{-1}$

$g'' = ?$

$$g'' = \frac{R^2}{(4R)^2} \times 9.81$$

$$= \frac{9.81}{16}$$

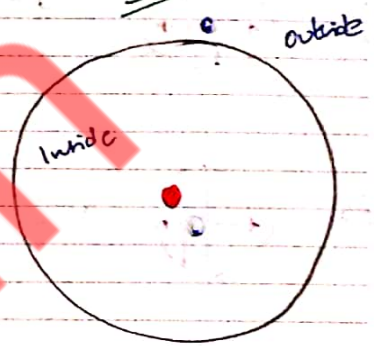
$$= 0.6\ Nkg^{-1}$$

Gravitational Field Strength

$$g = \frac{F}{m}$$ → general expressive

OR

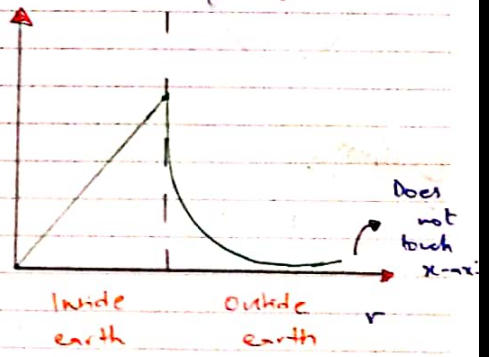
$$g = \frac{-GM}{r^2}$$ → For earth



Justification

Only valid for outside earth

$$g = \frac{GM}{r^2}$$ — only consider magnitude no -ve sign



* GM are constant

$$g \propto \frac{1}{r^2}$$ → It follows inverse square law

Inside earth

M does not remain constant ∴ $g = \frac{GM}{r^2}$ isn't applicable

(next page)

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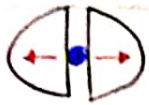
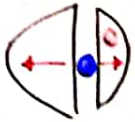
1. 

$g = \frac{F}{m}$

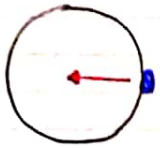
At unit mass from earth is same at 1 so m > 0

$\therefore = 0$

2. 

$g = \frac{F}{m} \neq 0$

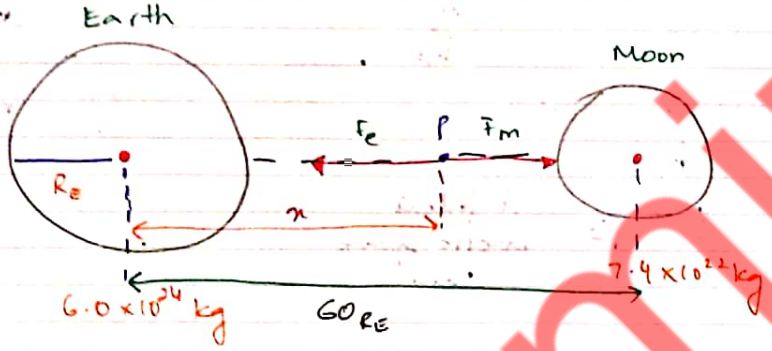
As there is a difference in unit mass so gravity will not $\neq 0$

3. 

$g = \frac{F}{m}$

max as m is maximum (whole of Earth)

Question



a) At point P => g = 0 (why)

Ans $\frac{F}{m} = 0$

$F = 0 \Rightarrow F_e = F_m$

F_e: force exerted by earth

F_m: force exerted by moon

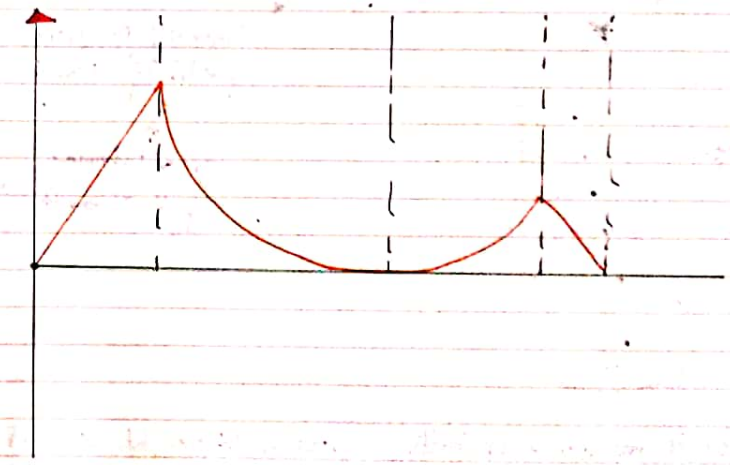
$$\frac{GM_e m}{x^2} = \frac{GM_m m}{(60R_e - x)^2}$$

$$\frac{M_e}{M_m} = \frac{x^2}{(60R_E - x)^2}$$

$$\sqrt{\frac{M_e}{M_m}} = \frac{x}{60R_E - x}$$

$$\boxed{x = 54R_E}$$

c)



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9-8-21 Monday

Gravitational Motion

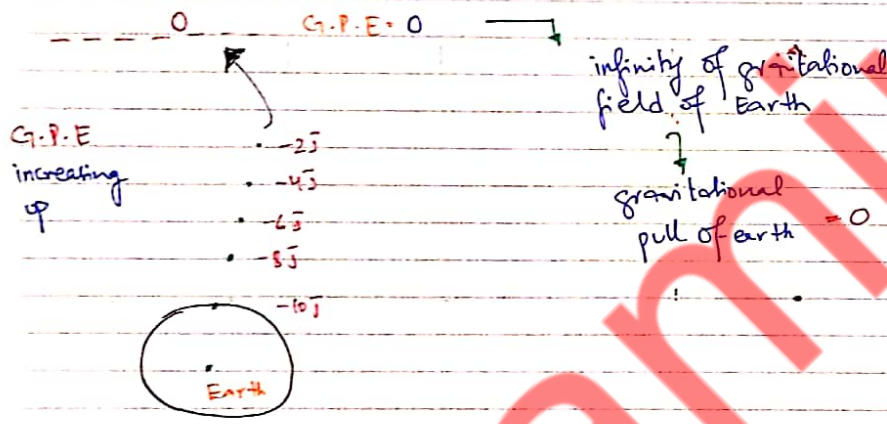
Gravitational potential energy

gravitational → force of gravity
potential → ability
energy → work done

$E_p = mgh$ g = gravitational field strength

→ g is not constant at every height, expression can only be used for those change in heights in which g is "constant"

→ Ability to do work under the force of gravity



infinity of gravitational field of Earth
↓
gravitational pull of earth = 0

* Ability to do work on hitting ground ∝ Height ∝ G.P.E

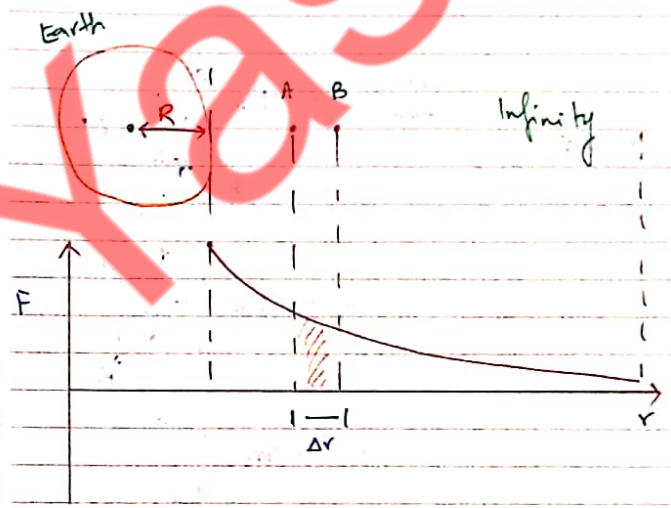
→ Height is max at infinity, G.P.E should be max. Max value is "0". At below infinity, all values of G.P.E are "-ve" → represent deficiency of energy in order to reach infinity

→ [illegible] energy required to reach infinity

-6J G.P.E ⇒ 6J more needed to reach infinity

→ At infinity, more energy is not needed as it's at infinity so "0".

→ Minimum energy at surface of Earth (outside earth)



→ curve following the inverse square law outside earth does not touch the x-axis

Area of graph = Work done
A & B are very close to each other
Δr = small distance b/w points

Area b/w A & B (shaded) = Work done b/w points A & B

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Work done = F × Δr

F = gravitational force

$$F = \frac{-GMm}{r^2}$$

$$F \times \Delta r = \boxed{\frac{-GMm}{r^2} \times \Delta r}$$

$$= \int_R^\infty \frac{-GMm}{r^2} \times \Delta r$$

$$\text{Work done} = -GMm \int_R^\infty \frac{1}{r^2} \times \Delta r$$

it is

$$= -GMm \left| -\frac{1}{r} \right|_R^\infty$$

$$= GMm \left| \frac{1}{r} \right|_R^\infty$$

is zero

$$\text{Work done} = GMm \left[\frac{1}{\infty} - \frac{1}{R} \right]$$

$$= GMm \left[0 - \frac{1}{R} \right]$$

$$= -\frac{GMm}{R}$$

→ This G.P.E at surface of Earth

→ work done from infinity to surface of earth

→ Δ in G.P.E from infinity to surface of earth

* Work done from infinity to surface of Earth represents the G.P.E at the surface of Earth

Gravitational Potential Energy

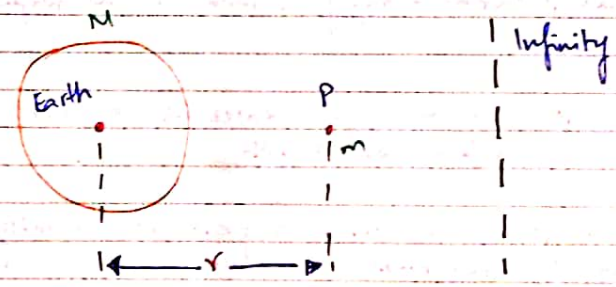
Work done in moving an object from infinity to a point inside the gravitational field is called "gravitational potential energy of object" at that point.

→ To find out any object's g.p.e inside earth's gravitational field so we will find out the work done to bring the object from infinity to the object's position, representing it's G.P.E.

$$\boxed{G.P.E = \frac{-GMm}{r}}$$

r = distance of object from the centre of earth

↳ put -ve sign as a part of magnitude



→ Work done to move P from infinity at m represents the g.p.e of m.

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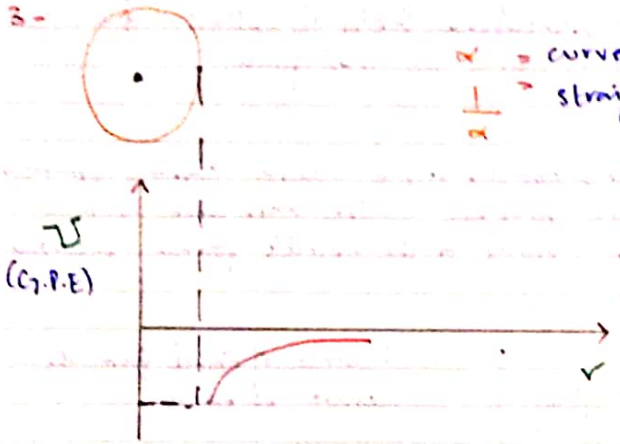
Properties

1- Unit = Joule

2- Scalar quantity — take -ve as part of magnitude

3-



$\propto$ = curved graph

$\frac{1}{\propto}$ = straight line graph

* Do not touch x-axis (as it'll only be zero in infinity)

i- Gravitational potential energy is minimum at the surface of earth

ii- Gravitational potential energy increases as we move away from the surface of Earth

iii- Gravitational potential energy is zero at infinity and it is at it's maximum value.

4- Gradient, $\frac{\Delta U}{\Delta r}$

$= \frac{U}{r}$

$= \frac{-GMm}{r} \div r$

$\Rightarrow \boxed{\frac{-GMm}{r^2}}$

↳ Force

G.P.E vs r

"Gradient of graph represents gravitational force"

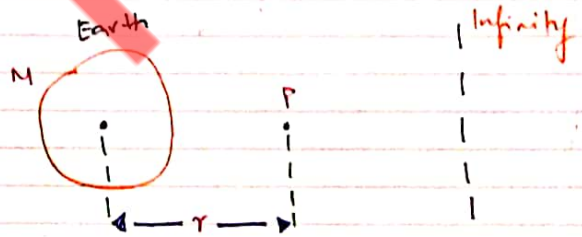
Gravitational Potential

G.P.E of unit mass at a point inside gravitational field is called "Gravitational potential at that point"

OR

Work done in moving unit mass from infinity to a point inside the gravitational field is called "Gravitational potential at that point".

→ G.P is $\frac{G.P.E}{mass}$ (the only difference of G.P.E and G.P is unit mass)



$$\boxed{\text{Gravitational potential} = \frac{G.P.E}{mass}}$$

$$\boxed{\phi = \frac{U}{m}}$$

At earth,

$$U = \frac{-GMm}{r}$$

$$\frac{\frac{-GMm}{r}}{m}$$

G = Universal Gravitational Constant

M = Mass of Earth

r = Dist-nce of point from earth

↓

where G.P are we are finding

At earth, gravitational potential

$$\boxed{U = \frac{-GM}{r}}$$

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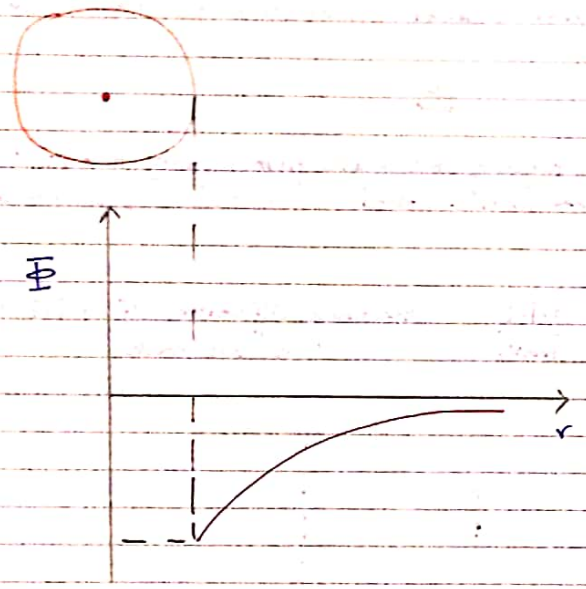
Properties

1- Unit => Jkg^{-1}

2- Scalar Quantity (don't prove -ve in magnitude)

3-



i- Gravitational potential is minimum at the surface of Earth

ii- Gravitational potential increases as we move away from the surface of Earth

iii- Gravitational potential is zero at infinity and it is at it's maximum value

$$\frac{\Delta \Phi}{\Delta r} = \frac{\Phi}{r} = -\frac{GM}{r} \div r = -\frac{GM}{r} \times \frac{1}{r} = \boxed{-\frac{GM}{r^2}} \text{ — } g$$

Gravitational field strength

* Gradient of Φ vs r graph will represent "gravitational field strength"

Q.1

a) M = ?

$g = \frac{GM}{r^2}$

$r \Rightarrow \frac{1.2 \times 10^4}{2} = 0.6 \times 10^4$ km $\times 10^3 \rightarrow$ into "m"

$W = X \cdot m\Phi_X$

$Y \cdot W = m\Phi_Y$

Planet

$8.6 = \frac{6.67 \times M}{(0.6 \times 10^4 \times 10^3)^2}$

$M = \frac{8.6 \times (0.6 \times 10^7)^2}{6.67 \times 10^{-11}}$

$M = \boxed{4.64 \times 10^{24} \text{ kg}}$

diameter = 1.2×10^4 km

$g = 8.6 \, Nkg^{-1}$

$G = 6.67 \times 10^{-11} \, Nm^2/kg^2$

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b) X – -5.3×10^{7} J/kg
Y – -6.8×10^{7} J/kg

i – Which point is nearer to the planet

Y is

Smaller G.P.E / G.P

* As we move away from planet the G.P or G.P.E increased
* For scalar quantities bigger minus smaller

V = ?

ii) If an object drops from X to Y, it gains speed therefor K.E increases and potential energy decreases

Gain in E_K = loss in E_p

$\frac{1}{2}mv^2 - 0 = U_X - U_Y$

$\frac{1}{2}mv^2 = m\Phi_X - m\Phi_Y$

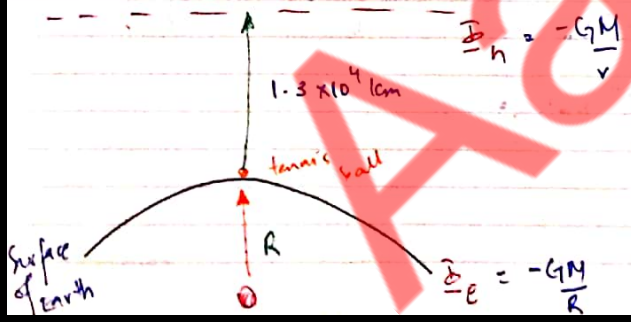
$\frac{1}{2}\cancel{m}v^2 = \cancel{m}[\Phi_X - \Phi_Y]$

$\frac{1}{2}v^2 = [(-5.3\times10^7) - (-6.8\times10^7)]$

$v^2 = \boxed{5.3\times10^{+3}\ ms^{-1}}$

$R = 6.4\times10^6$ m
$M = 6.0\times10^{24}$ kg
$G = 6.67\times10^{-11}$ Nm²/kg²

Q.2



$\Phi_h = -\frac{GM}{r}$ — distance b/w earth's surface to the Φ_h)

$\Phi_E = -\frac{GM}{R}$

$\Delta\Phi = \Phi_h - \Phi_E$

$\Rightarrow (-3.28\times10^7) - (-6.25\times10^7)$

$\Rightarrow 2.97\times10^7$ J/kg⁻¹

b) V = ?

When object moves from surface of Earth to h

Loss in E_k = Gain in E_p

$\frac{1}{2}\cancel{m}v^2 = \cancel{m} \times \Delta\Phi$

$\frac{1}{2}v^2 = \Delta\Phi$

$v^2 = 2 \times 2.97\times10^7$

$\sqrt{v^2} = \sqrt{59400000}$

$v = \boxed{7.7\times10^3\ m/s}$

Thrown from surface of earth to a certain height

* At surface of earth $\Rightarrow -\frac{GM}{R}$

At surface of earth $= \frac{-6.67\times10^{-11}\times6.0\times10^{24}}{6.4\times10^6}$

$= -6.25\times10^7$

At h $\Rightarrow -\frac{GM}{r}$

$= \frac{-6.67\times10^{-11}\times6.4\times10^{24}}{(1.3\times10^4\times10^3)}$

$= -3.28\times10^7$

$\Phi = \frac{U}{m}$

$U = \Phi m$

$\Delta U = \Delta\Phi \times m$

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Q.3

$$\Delta E_p = mg(\Delta h)$$

Prove this expression

$g = \frac{GM}{r^2}$

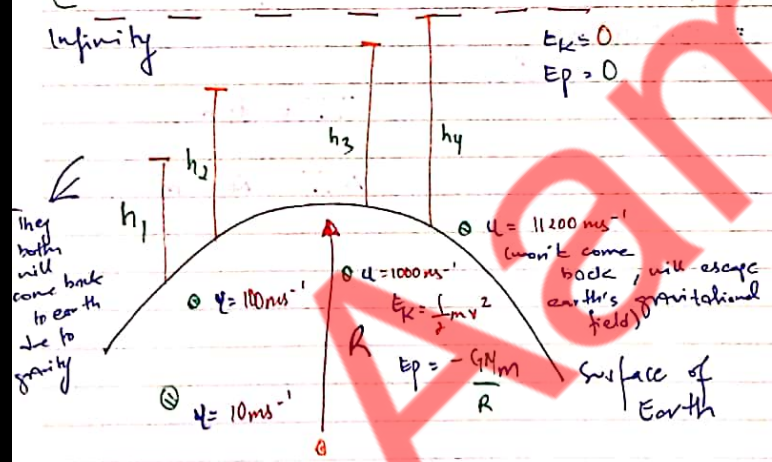
* Δh — small change in height

as for Δh "g" remains constant

Force of gravity = 0
G.P or G.P.E = 0/max

Infinity

$E_K = 0$
$E_p = 0$



* As we increase the speed of projection, the height attained by tennis ball increases

Minimum speed required to throw an object out of gravitational field is called "Escape speed of gravitational field"

→ It is associated with the gravitational field, every planet or star has different escape speed.

$$\text{Escape Speed} = \sqrt{\frac{2GM}{R}}$$

M = Mass of planet
G = Universal gravitational constant
R = Radius of planet

for earth it's = 11200 ms⁻¹

→ Initial speed is u but we assume that it only helps the ball to reach infinity — v = 0

u = escape speed

Loss in E_K = Gain in E_p

$$\frac{1}{2}mu^2 - 0 = 0 - \left(-\frac{GMm}{R}\right)$$

$$\frac{1}{2}\cancel{m}u^2 = \frac{GM\cancel{m}}{R}$$

$$u^2 = \frac{2GM}{R}$$

$$u = \sqrt{\frac{2GM}{R}}$$

→ Object's mass does not matter if it needs to be escaping the earth's gravitational field

→ u needs to be 11,200 ms⁻¹ as it doesn't depend on force.

Avg speed of → Air molecules is less than earth's escape speed but more than moon's escape speed

It is the reason why the has Atmosphere

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Few stars are so dense that their escape speed are even more than the speed of light.

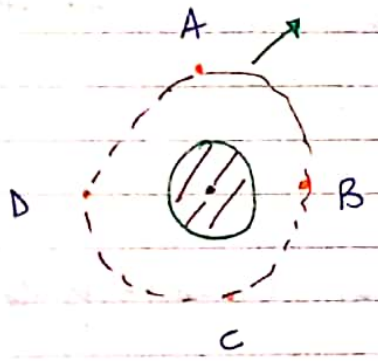
↓

So stars appear dark => Black hole

$$\Phi = -\frac{GM}{r}$$

r — distance of object from centre of earth

At this all points have same Ep / G.P



→ All these 4 points are at equal distance from centre of Earth

So same G.P

∴ Equipotential points

Equipotential Points

Points having same distance within the gravitational field from the centre of object.

OR

Points having same potential within the gravitational field are called "Equipotential Points"

Equipotential Surface

A surface passing through all those points which have same potential within the gravitational field is called "Equipotential surface"

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